📊 1. General Sales Analysis

Business Question: What is the total revenue generated, how many orders were made in 2025, and how do sales evolve throughout the year?

🔹 SQL – Total revenue generated:

SQL

```
SELECT SUM(revenue) AS total_revenue
FROM synthetic_ecommerce_sales;
```

Result:

Total Revenue
24,698,636.54

🔹 SQL – Total number of orders in 2025:

SQL

```
SELECT COUNT(order_id) AS total_orders
FROM synthetic_ecommerce_sales
WHERE order_date >= '2025-01-01'
AND order_date < '2026-01-01';
```

Result:

Total Orders
33,600

🔹 SQL – Monthly sales evolution:

SQL

```
SELECT strftime('%Y-%m', order_date) AS month,
    COUNT(order_id) AS total_orders,
    SUM(product_price * quantity) AS total_revenue
FROM synthetic_ecommerce_sales
GROUP BY month
ORDER BY month;
```

Result:

Month	Total Orders	Total Revenue
2025-01	2,864	2,053,229.85
2025-02	2,585	1,888,985.25
2025-03	2,809	2,055,598.05
2025-04	2,779	2,016,263.97
2025-05	2,906	2,068,441.36
2025-06	2,750	1,953,417.71
2025-07	2,792	2,033,989.93
2025-08	2,873	2,092,621.35
2025-09	2,730	1,949,839.88
2025-10	2,886	2,046,029.51
2025-11	2,774	2,456,581.85
2025-12	2,852	2,083,637.83

🧠 **Insight:** Sales show seasonal patterns, with notable peaks in November and August, likely associated with promotions or special events. Although total orders and revenue remain relatively stable month-to-month, the peaks indicate strategic moments to boost sales campaigns.

🎯 **Conclusion:** The business has a consistent sales flow throughout the year, with opportunities to optimize promotions and marketing during peaks to maximize revenue. **Recommendation:** Strategically plan campaigns and stock for months with higher demand (November and August), ensuring product availability and fast deliveries to increase customer satisfaction and sales.

🛍️ 2. Product Category Analysis

Business Question: Which product categories generate the most revenue and how many units are sold in each?

🔹 SQL – Total revenue by category:

SQL

```
SELECT product_category,
    SUM(product_price * quantity) AS revenue_total
```

```
FROM synthetic_ecommerce_sales
GROUP BY product_category
ORDER BY revenue_total ASC;
```

Result:

Category	Revenue
Sports	3,747,241.70
Home	3,746,857.48
Beauty	3,733,643.30
Automotive	3,724,347.36
Electronics	3,707,297.85
Fashion	3,677,375.29
Toys	3,670,973.91

🔹 SQL – Units sold by category:

SQL

```
SELECT product_category,
    SUM(quantity) AS units_sold
FROM synthetic_ecommerce_sales
GROUP BY product_category
ORDER BY units_sold DESC;
```

Result:

Category	Units
Beauty	15,043
Home	14,923
Sports	14,892
Electronics	14,874
Fashion	14,802
Automotive	14,719
Toys	14,567

🔹 SQL – Most profitable categories:

```
SQL
SELECT region,
    SUM(revenue) AS total_revenue
FROM synthetic_ecommerce_sales
GROUP BY region
ORDER BY total_revenue ASC;
```

Result:

Category	Revenue
Sports	3,747,241.70

🧠 **Insight:** Revenue and sales volume are relatively balanced across all categories, indicating that the business does not depend on a single category. Beauty, Home, and Sports lead slightly in both income and units sold, while Toys and Fashion lag slightly behind. This suggests that while the portfolio is diversified, certain categories show greater growth potential if focused strategies are applied.

🎯 **Conclusion:** The business has a balanced portfolio, which reduces concentration risks. **Recommendation:** Boost lower-performing categories (Toys and Fashion) through promotions, marketing, or price adjustments, and replicate successful strategies from leading categories (Beauty, Home, Sports) to maximize income and sales volume.

🌎 3. Geographic Analysis

Business Question: What is the revenue by region, how many orders are generated in each, and which are the strongest and weakest markets?

🔹 SQL – Revenue by region:

```
SQL
SELECT region,
    SUM(revenue) AS total_revenue
FROM synthetic_ecommerce_sales
GROUP BY region
ORDER BY total_revenue ASC;
```

Result:

Region	Total Revenue

North America	4,054,954.19
Africa	4,064,644.95
South America	4,095,811.34
Asia	4,121,620.67
Europe	4,143,562.44
Oceania	4,218,042.95

🔹 SQL – Number of orders by region:

SQL
```
SELECT region,
    COUNT(order_id) AS total_orders
FROM synthetic_ecommerce_sales
GROUP BY region
ORDER BY total_orders ASC;
```

Result:

Region	Total Orders
Africa	5,547
North America	5,566
Europe	5,570
Asia	5,588
South America	5,610
Oceania	5,719

🔹 SQL – Identify strongest and weakest markets:

SQL

```
-- Strongest market (highest revenue)

SELECT region, SUM(revenue) AS total_revenue
FROM synthetic_ecommerce_sales
GROUP BY region
ORDER BY total_revenue DESC
LIMIT 1;

-- Weakest market (lowest revenue)
```

```
SELECT region, SUM(revenue) AS total_revenue
FROM synthetic_ecommerce_sales
GROUP BY region
ORDER BY total_revenue ASC
LIMIT 1;
```

Result:

Tipo de mercado	Región	Revenue
Más fuerte	Oceania	4,218,042.95
Más débil	North America	4,054,954.19

🧠 **Insight:** Oceania and Europe are the markets with the highest revenue, while North America and Africa show lower relative performance. The distribution of orders follows a similar pattern to revenue, indicating that differences are not due to a few large orders but to consistency in sales. This demonstrates that certain regions present growth opportunities, while others might require market optimization or more aggressive marketing strategies.

🎯 **Conclusion:** The business has geographically heterogeneous markets, with leading and lagging regions. **Recommendation:** Implement personalized regional strategies focused on increasing sales in less profitable markets and consolidating presence in strong markets through promotions, loyalty campaigns, or logistical adjustments.

💳 4. Payment Methods

Business Question: What is the distribution of orders and revenue by payment method, and which method is associated with higher average purchases?

🔹 SQL – Order distribution by payment method:

```
SQL
SELECT payment_method,
COUNT(order_id) AS total_orders
FROM synthetic_ecommerce_sales
GROUP BY payment_method
ORDER BY total_orders ASC;
```

Result: Distribución de órdenes

Payment Method	Total Orders
Cash	8,302

BankTransfer	8,381
PayPal	8,394
CreditCard	8,523

🔹 SQL – Revenue generated by each payment method:

SQL
```
SELECT payment_method,
    SUM(revenue) AS total_revenue
FROM synthetic_ecommerce_sales
GROUP BY payment_method
ORDER BY total_revenue ASC;
```

Result:

Payment Method	Total Revenue
BankTransfer	6,127,846.84
PayPal	6,143,359.38
Cash	6,209,751.09
CreditCard	6,217,679.23

🔹 SQL – Average revenue per order by payment method:

SQL

```
SELECT payment_method,
    COUNT(order_id) AS total_orders,
    SUM(revenue) AS total_revenue,
    ROUND(AVG(revenue), 2) AS average_revenue_per_order
FROM synthetic_ecommerce_sales
GROUP BY payment_method
ORDER BY average_revenue_per_order DESC;
```

Result: Revenue promedio por orden

Payment Method	Total Orders	Total Revenue	Average Revenue per Order
Cash	8,302	6,209,751.09	747.98
PayPal	8,394	6,143,359.38	731.88
BankTransfer	8,381	6,127,846.84	731.16
CreditCard	8,523	6,217,679.23	729.52

🧠 **Insight:** Although CreditCard records the highest number of orders and Cash a slightly lower volume, the highest average revenue per order belongs to Cash, followed by PayPal. This indicates that customers paying in cash tend to make larger purchases, while digital methods have more volume but lower average tickets.

🎯 **Conclusion:** The business can leverage purchasing behavior by payment method to design personalized marketing strategies and promotions:

- Incentivize Cash or PayPal payments for high-value products.
- Optimize campaigns targeted at CreditCard users to increase purchase frequency, even with lower tickets.

🚚 5. Logistics and Deliveries

Business Question: What is the average delivery time, how does it relate to customer satisfaction and returns, and do slower deliveries affect the return rate?

🔹 SQL – Average delivery days

SQL
```
SELECT ROUND(AVG(delivery_days), 2) AS average_delivery_days
FROM synthetic_ecommerce_sales;
```

Result:

Métrica	Valor
Promedio delivery_days	4.99

🔹 SQL – Relationship between delivery days and customer rating

SQL
```
SELECT delivery_days,
    ROUND(AVG(customer_rating), 2) AS avg_rating,
    COUNT(order_id) AS total_orders
FROM synthetic_ecommerce_sales
GROUP BY delivery_days
ORDER BY delivery_days;
```

Result:

Delivery Days	Avg Rating	Total Orders
1	3.51	3,774

2	3.50	3,787
3	3.51	3,712
4	3.49	3,731
5	3.52	3,674
6	3.48	3,812
7	3.52	3,638
8	3.52	3,674
9	3.48	3,798

🔹 SQL – Relationship between delivery days and returns

SQL
```
SELECT delivery_days,
    COUNT(order_id) AS total_orders,
    SUM(is_returned) AS total_returns,
    ROUND(100.0 * SUM(is_returned) / COUNT(order_id), 2) AS return_rate_percent
FROM synthetic_ecommerce_sales
GROUP BY delivery_days
ORDER BY delivery_days;
```

Result:

Delivery Days	Total Orders	Total Returns	Return Rate (%)
1	3,774	232	6.15
2	3,787	235	6.21
3	3,712	233	6.28
4	3,731	224	6.00
5	3,674	237	6.45
6	3,812	232	6.09
7	3,638	223	6.13
8	3,674	204	5.55

🧠 **Insight:** The average delivery time is approximately 5 days. Customer ratings remain fairly constant (3.48–3.52), indicating that customers do not show a significant impact on satisfaction due to small variations in delivery speed. The return rate is also quite stable,

ranging between 5.55% and 6.45%, and there is no clear increase in returns for slower deliveries.

🎯 **Conclusion:** Logistics performance is consistent and does not generate a significant negative impact on customer experience or the return rate. **Recommendation:** Maintain current delivery processes and continuously monitor logistical efficiency, but modifying delivery times based on this data is not a priority.

🔁 6. Returns

Business Question: What is the overall return rate, which categories and regions have the highest returns, and how much does this impact revenue?

🔹 SQL – Overall return rate

SQL
```
SELECT
  COUNT(order_id) AS total_orders,
  SUM(is_returned) AS total_returns,
  ROUND(100.0 * SUM(is_returned) / COUNT(order_id), 2) AS return_rate_percent
FROM synthetic_ecommerce_sales;
```

Result: | Total Orders | Total Returns | Return Rate (%) | | :--- | :--- | :--- | | 33,600 | 2,038 | 6.07 |

🔹 SQL – Categories with the most returns

SQL
```
SELECT product_category,
    SUM(is_returned) AS total_returns,
    COUNT(order_id) AS total_orders,
    ROUND(100.0 * SUM(is_returned) / COUNT(order_id), 2) AS return_rate_percent
FROM synthetic_ecommerce_sales
GROUP BY product_category
ORDER BY total_returns DESC;
```

Result:

Product Category	Total Returns	Total Orders	Return Rate (%)
Fashion	556	4,803	11.58
Automotive	272	4,712	5.77

Home	262	4,846	5.41
Electronics	256	4,810	5.32
Toys	234	4,768	4.91
Beauty	230	4,859	4.73
Sports	228	4,802	4.75

🔹 SQL – Regions with the highest return rate

SQL
```
SELECT region,
    COUNT(order_id) AS total_orders,
    SUM(is_returned) AS total_returns,
    ROUND(100.0 * SUM(is_returned) / COUNT(order_id), 2) AS return_rate_percent
FROM synthetic_ecommerce_sales
GROUP BY region
ORDER BY return_rate_percent DESC;
```

Result:

Region	Total Orders	Total Returns	Return Rate (%)
North America	5,566	372	6.68
Asia	5,588	351	6.28
South America	5,610	343	6.11
Oceania	5,719	335	5.86
Africa	5,547	321	5.79
Europe	5,570	316	5.67

🔹 SQL – Impact of returns on revenue

SQL
```
SELECT SUM(revenue) AS total_revenue,
    SUM(CASE WHEN is_returned = 1 THEN revenue ELSE 0 END) AS
revenue_lost_due_to_returns,
    ROUND(100.0 * SUM(CASE WHEN is_returned = 1 THEN revenue ELSE 0 END) /
SUM(revenue), 2) AS percent_lost
FROM synthetic_ecommerce_sales;
```

Result:

Total Revenue	Revenue Lost due to Returns	Percent Lost (%)
24,698,636.54	1,504,965.93	6.09

🧠 **Insight:** The overall return rate is 6.07%, which is relatively low. The Fashion category accounts for the highest proportion of returns (11.58%), while other categories show more balanced rates (~4.7–5.8%). Geographically, North America and Asia show a slightly higher return rate than other regions. The financial impact of returns represents approximately 6% of total revenue, indicating that returns are a relevant but not critical factor.

🎯 **Conclusion:** Although the return rate is moderate, the Fashion category requires special attention to reduce returns, possibly through improved product descriptions, sizing, or clearer return policies. Quality and logistics strategies could also be applied to other categories to minimize losses.

⭐ 7. Customer Satisfaction

Business Question: What is the average level of customer satisfaction and how does it vary by product price range?

🔹 SQL – Average customer rating

SQL
```
SELECT
  ROUND(AVG(customer_rating), 2) AS average_rating
FROM synthetic_ecommerce_sales;
```

Result:

Average Rating
3.50

🔹 SQL – Relationship between rating and price range

SQL
```
SELECT
  CASE
    WHEN product_price < 50 THEN '<50'
    WHEN product_price BETWEEN 50 AND 100 THEN '50-100'
    WHEN product_price BETWEEN 101 AND 200 THEN '101-200'
    ELSE '>200'
  END AS price_range,
  ROUND(AVG(customer_rating), 2) AS avg_rating,
```

```
  COUNT(order_id) AS total_orders
FROM synthetic_ecommerce_sales
GROUP BY price_range
ORDER BY price_range;
```

Result:

Price Range	Average Rating	Total Orders
101-200	3.50	6,688
50-100	3.50	3,362
<50	3.48	3,103
>200	3.51	20,447

🧠 **Insight:** The overall average satisfaction is 3.5/5, showing moderate satisfaction among most customers. Products priced >200 have the highest rating (3.51), while the cheapest ones (<50) receive a slightly lower valuation (3.48). This reflects that customers perceive higher value or quality in more expensive products, while mid-ranges maintain stable satisfaction.

🎯 **Conclusion:** It is recommended to focus experience and communication strategies on cheaper products to raise their perception. Additionally, leveraging the high satisfaction in premium products (>200) through loyalty programs or upselling could maximize retention and revenue.

🏷️ 8. Impact of Discounts

Business Question: How do discounts impact revenue and order quantity? Do they increase sales or reduce profitability?

🔹 SQL – Average discount applied

SQL
```
SELECT
  ROUND(AVG(discount_percent), 2) AS average_discount
FROM synthetic_ecommerce_sales;
```

Result:

Average Discount (%)
5.03

🔹 SQL – Revenue with vs. without discount

```
SQL
SELECT
  CASE
    WHEN discount_percent = 0 THEN 'No Discount'
    ELSE 'With Discount'
  END AS discount_type,
  SUM(revenue) AS total_revenue,
  COUNT(order_id) AS total_orders
FROM synthetic_ecommerce_sales
GROUP BY discount_type;
```

Result:

Discount Type	Total Revenue	Total Orders
No Discount	12,999,473.39	16,820
With Discount	11,699,163.15	16,780

🔹 SQL – Profitability impact evaluation

```
SQL
SELECT
  CASE
    WHEN discount_percent = 0 THEN 'No Discount'
    ELSE 'With Discount'
  END AS discount_type,
  COUNT(order_id) AS total_orders,
  SUM(revenue) AS total_revenue,
  ROUND(AVG(revenue), 2) AS average_revenue_per_order
FROM synthetic_ecommerce_sales
GROUP BY discount_type;
```

Result:

Discount Type	Total Orders	Total Revenue	Average Revenue per Order
No Discount	16,820	12,999,473.39	772.86
With Discount	16,780	11,699,163.15	697.21

🧠 **Insight:** Although discounts cover nearly half of the orders, total revenue and average revenue per order are clearly higher for products without discounts. This indicates that while discounts attract sales, they reduce profitability per order.

🎯 **Conclusion:** It is recommended to use discounts strategically for products that need turnover or to attract new customers, but avoid applying them globally as they reduce average income per sale and affect profitability.

📈 9. Temporal Analysis

Business Question: How does the business evolve throughout the year 2025? Which months show the highest sales and revenue peaks?

🔹 SQL – Monthly trends by date

```
SQL
SELECT
  strftime('%Y-%m', order_date) AS month,
  COUNT(order_id) AS total_orders,
  SUM(revenue) AS total_revenue
FROM synthetic_ecommerce_sales
GROUP BY month
ORDER BY month;
```

Result:

Month	Total Orders	Total Revenue
2025-01	2,864	2,053,229.85
2025-02	2,585	1,888,985.25
2025-03	2,809	2,055,598.05
2025-04	2,779	2,016,263.97
2025-05	2,906	2,068,441.36
2025-06	2,750	1,953,417.71
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2025-08	2,873	2,092,621.35
2025-09	2,730	1,949,839.88
2025-10	2,886	2,046,029.51
2025-11	2,774	2,456,581.85
2025-12	2,852	2,083,637.83

🔹 SQL – Detecting sales peaks

```
SQL
SELECT
  strftime('%Y-%m', order_date) AS month,
  COUNT(order_id) AS total_orders,
  SUM(revenue) AS total_revenue
FROM synthetic_ecommerce_sales
GROUP BY month
ORDER BY total_revenue DESC
LIMIT 5;
```

Result:

Month	Total Orders	Total Revenue
2025-11	2,774	2,456,581.85
2025-08	2,873	2,092,621.35
2025-12	2,852	2,083,637.83
2025-05	2,906	2,068,441.36
2025-03	2,809	2,055,598.05

🧠 **Insight:** The business shows stable evolution throughout the year, with clear peaks in November and August, which could be related to seasonal campaigns or special promotions. March and May also show high revenues, indicating months of strong but more moderate demand.

🎯 **Conclusion:** It is recommended to plan inventory and marketing in advance for months with sales peaks, especially November and August, to maximize revenue. Additionally, analyzing the factors driving these peaks can help replicate successful strategies in other months.